

What Makes a Scientific Question Succeed?

Predicting Future Attention, Resolution, and Premise Revision from Question Structure and Evidence Context

Paper 3 — temporally grounded predictor discovery for scientific questions

Abstract

Large language models are increasingly asked to propose “important open questions”, and are typically evaluated by asking another model how good those questions feel. We replace taste with history. We construct a temporally grounded dataset of **980 astronomy research questions** frozen at five historical cutoffs (2012–2020), spanning eight subfields and four distinct generation sources — evidence-tension mining, direct LLM elicitation, author-stated future work, and weak/negative controls — on top of the complete arXiv astro-ph metadata record (**313,189 papers, 2005–2025**). Each question carries a six-group cutoff-time feature record (text structure, evidence context, novelty, falsifiability, tractability, field environment) and a tiered outcome label derived from the five years of literature that actually followed. Under a strict, source-blinded outcome judge, 52.7% of questions were substantively addressed; negative-control questions were addressed least (32%) and direct-LLM questions most (88%), the latter carrying a documented parametric-leakage caveat. Interpretable models trained on 2012–2016 cutoffs predict 2020 outcomes at AUC 0.71 and transfer to entirely held-out subfields at AUC 0.65–0.71. Across univariate, controlled, ablation, and stability analyses, the robust predictors of future attention are **prior community recognition** (overlap with pre-cutoff review articles, $\beta = +0.34$, $p < 10^{-4}$), **the density of directly related prior evidence** ($\beta = +0.26$), and **explicitly stated competing hypotheses** ($\beta = +0.17$) — while high entity-specificity predicts *less* and *slower* engagement under literature-bounded labels. Five of six pre-stated mechanism hypotheses, including “evidence tension beats novelty”, are *not* supported at scale — a result only visible because the dataset is large, multi-cutoff, and control-laden. All data, code, labels, and judge rationales are released.

1 Introduction

A scientific question is an instrument for allocating future effort. Yet the standard way of evaluating machine-generated research questions — LLM or expert panels scoring “importance” and “novelty” at generation time — measures how a question *sounds*, not what it *does*. Paper 2 of this series introduced historical backtesting as the alternative: freeze questions using only pre-cutoff literature, then observe what the scientific community actually did afterwards. Its pilot instance, however, contained ten questions from a single subfield and cutoff — enough for a case study, not for learning what distinguishes questions that succeed.

This paper turns the backtesting protocol into a supervised-learning problem at scale. Three design decisions matter:

1. **The unit of data is (question, cutoff, subfield, source), not the question alone.** We generate questions at five rolling cutoffs (2012–2020) in eight astronomy subfields, so that one study contains what would otherwise be forty historical experiments, and era- or field-specific accidents can be detected rather than absorbed.
2. **No single generator.** Predictors learned from one generation system risk being that system’s stylistic fingerprint. We use four sources — our evidence-tension system, a direct-LLM baseline, questions extracted from the authors’ own future-work statements, and

deliberately weak controls (random claim pairings, vague, untestable, and evidence-free-contrarian questions). Without negative controls a model can only learn that “question-shaped sentences succeed”.

3. **Prediction is out-of-time or out-of-domain, always.** Random splits would place near-duplicate questions from the same era and topic on both sides. We train on 2012–2016 cutoffs, validate on 2018, test on 2020, and additionally hold out entire subfields.

The contribution is deliberately not a leaderboard number. It is a temporally grounded dataset of scientific questions with subsequent outcomes, and an analysis of *which properties of a question and its evidence context* predicted future scientific attention, resolution, and premise revision — with controls for field popularity, instrument eras, and generation method, and with stability checks across cutoffs, subfields, and sources.

2 Related framing

Paper 1 built an evidence-graph system that generates questions from cross-paper observational tensions. Paper 2 defined the backtesting benchmark and showed, on ten frozen exoplanet-atmosphere questions, that future literature substantively engaged all ten (one premise later refuted). Both left open the question this paper addresses: *what, measurably, makes a question one the community will take up?* Related work on predicting scientific impact operates at the paper or grant level (citation prediction, idea novelty metrics); to our knowledge no prior dataset attaches future-outcome labels to *questions* frozen at multiple historical cutoffs with controlled generation sources.

3 Dataset construction

3.1 Corpus

We harvested the complete arXiv astro-ph metadata record — titles, abstracts, submission dates, and categories — from 2005-01 through 2025-12 via the public arXiv API (**313,189 papers** after deduplication; `pipeline/harvest.py`). Papers are assigned to eight subfields by transparent weighted keyword rules over title+abstract (multi-membership allowed): exoplanet atmospheres (4,354 papers), protoplanetary disks (10,378), stellar activity (4,347), fast radio bursts (1,966), gravitational waves (17,989), galaxy evolution (21,390), cosmology tensions (16,403), and compact objects (34,266). The classifier is identical at every cutoff, so subfield corpora are time-sliced views of a fixed rule, not retrofitted topic models.

Each cutoff year $c \in \{2012, 2014, 2016, 2018, 2020\}$ defines a **past window** $[c - 7, c]$ used for generation and features and a **future window** $(c, c + 5]$ used only for labels. The five future windows all close by 2025-12-31, giving every cutoff the same 5-year outcome horizon.

3.2 Question generation (40 cells $\times$ quota 25)

For every (cutoff, subfield) cell:

- **Evidence-tension (10).** We mine the cell’s past-window abstracts for tension statements (contradiction, discrepancy, unexplained-result markers), cluster them so that each cluster spans ≥ 2 distinct papers, and have an LLM phrase each cluster as one precise question grounded only in the mined sentences. Provenance (exact sentences, arXiv ids, dates) is stored with the question.

- **Direct LLM (5).** The baseline the field implicitly uses: the model sees a year-stratified sample of pre-cutoff titles and is asked for the most important unanswered questions, with an explicit knowledge-freeze instruction.
- **Future-work (5).** Sentences in which pre-cutoff authors themselves flag an open problem (“remains poorly constrained”, “further observations are needed”), selected for topic diversity and converted faithfully into interrogative form. These are real scientists’ questions.
- **Controls (5).** Two random claim-pairings across unrelated papers, one vague question, one untestable question, one evidence-free consensus challenge — all phrased by the same LLM so that surface style cannot separate controls from real candidates.

Near-duplicates within a cell are removed (TF-IDF cosine ≥ 0.75). The ten frozen Paper 1 evidence-graph questions join the dataset as a small additional source (cutoff 2020, exoplanet atmospheres), linking this study to the original system.

Realized totals: 980 questions — 374 evidence-tension, 200 direct-LLM, 196 future-work, 200 controls, 10 Paper 1. Every non-FRB cell filled its full quota of 25; early fast-radio-burst cells are data-limited by history itself (the pre-2013 FRB literature contains 21 papers, yielding 11 questions at the 2012 cutoff), which we treat as a feature of honest backtesting rather than a defect. The dataset is regenerable bit-identically from the frozen LLM cache; rebuilding the corpus from scratch and regenerating produced byte-identical questions.

Temporal isolation is enforced by construction — every mined sentence and every title shown to a generator predates the cutoff — and verified by an automated validator over the frozen records (980/980 pass).

3.3 Feature record (six groups, all cutoff-time)

- A. Text structure** length, named entities, numbers, causal / comparative / mechanistic language, measurable quantities, explicit alternatives (“real or artifact”), falsification wording, yes/no form, and a composite specificity score.
- B. Evidence context** retrieval of the top-20 pre-cutoff subfield papers for each question: similarity mass (top-1, mean top-5, count above threshold), instrument diversity among retrieved papers, density of tension markers in the retrieved evidence, and the age profile of that evidence; plus native evidence-record features (paper count, year spread) for tension-sourced questions.
- C. Novelty** nearest-neighbour semantic distance to the pre-cutoff corpus, mean top-10 similarity, concept-pair novelty (do the question’s two most distinctive terms co-occur in any pre-cutoff paper?), and overlap with pre-cutoff review articles (was the question already posed in reviews?).
- D. Falsifiability / answerability** rule-based markers plus a blinded LLM structured coding of {observable, competing hypotheses, quantitative test, falsification path} for every question.
- E. Tractability** facilities named in the question checked against a lexicon of ~ 50 astronomical facilities with operational epochs: counts of already-operational vs. future instruments, archival-data markers, longitudinal / theory / large-sample requirements.
- F. Field environment (controls)** subfield publication volume and growth at the cutoff, review activity, share of all astro-ph output, and proximity of the subfield’s next major facility launch (capped at 10 years). These are confounders to control for, not virtues of the question: without them a model mistakes hot fields for good questions.

3.4 Tiered outcome labels

Tier 1 (automatic candidates). For each question we retrieve future papers from its subfield’s future window: relevant-paper count (cosine ≥ 0.18), weak-relevance count, first-follow-up date, lead time in months, and review recognition.

Tier 2 (strict, blinded LLM judge). For every question with any plausible future evidence (956 of 980; the rest are auto-labeled `not_addressed`), a GPT-4o judge sees the question and its top-8 retrieved future records — never the generation source. The rubric is deliberately strict and two-stage: the judge first classifies every candidate paper as **direct** (investigates this question’s own objects, quantities, or claimed relationship), **adjacent** (same topic only), or unrelated, and only then assigns **addressed**, `answer_status` $\in \{\text{answered, partially_addressed, posed_but_open, premise_refuted, not_addressed}\}$, `premise_status`, supporting ids, a rationale, and a confidence. **addressed** is *derived*: it requires at least one direct candidate. This rubric matters — a first-pass lenient judge (GPT-4o-mini, same evidence) called 94% of questions addressed by accepting topical overlap; the strict per-candidate rubric brings the rate to 52.7% and restores variance to every analysis downstream.

Tier 3 (independent verification). A second model (GPT-4o-mini) re-judges all `premise_refuted` cases plus a 20% random sample under the identical rubric ($n = 200$): raw agreement on **addressed** is 63.5% ($\kappa = 0.21$), status agreement 58.4% ($\kappa = 0.22$). The disagreement is one-directional — the weaker model is systematically more lenient, almost never the reverse (its “partially_addressed” precision against the strict judge is 0.99) — so we treat the strict labels as primary and release both, with all rationales and supporting ids, for audit.

4 Models and evaluation protocol

Baselines before models, interpretable models before ensembles: majority class; field-popularity-only; question-text-only (bag of words); then L2 and L1 logistic regression, random forest, and gradient boosting on the structured features; multinomial GBM for the five-way status; a Cox proportional-hazards model for time-to-first-follow-up; and within-cell ranking concordance between model scores and future paper counts.

All headline numbers use the **temporal split** (train 2012–2016, validate 2018, test 2020). Generalization is probed with **leave-one-subfield-out** and a fixed **cross-domain split** (train: exoplanet atmospheres, disks, stellar activity, galaxy evolution $\rightarrow$ test: FRBs, gravitational waves, cosmology tensions, compact objects). Random splits are not reported anywhere in this paper.

Predictor identification goes beyond feature importance: (i) univariate associations; (ii) controlled logistic effects with subfield, cutoff, source, and environment covariates; (iii) feature-group ablations; (iv) sign-stability of controlled effects across cutoffs, subfields, and generation sources. A predictor is called robust only if it survives all four.

Pre-stated hypotheses

- H1** Cross-source evidence tension predicts future attention better than semantic novelty.
- H2** Observation availability predicts short-term attention but not premise refutation.
- H3** Explicit competing hypotheses raise the probability of being answered, given attention.
- H4** Novelty shows an inverted-U: very high novelty reduces short-term attention.
- H5** Field popularity inflates future paper counts without raising answer rates.
- H6** High-tension $\times$ high-tractability questions succeed most.

5 Results

5.1 What happened to the questions

Of 980 frozen questions, **52.7% were substantively addressed** within five years of their cutoff: 474 partially addressed, 30 posed-but-open, 8 answered, 4 premise-refuted, 464 not addressed. Addressed questions drew a mean of 3.8 relevant future papers and were first engaged after a mean of 16.7 months. Rates are stable across cutoffs (0.45–0.57), confirming that no single era drives the results.

Generation source separates outcomes exactly as a functioning label should (Table 1, Figure 1).

Table 1: Addressed rate by generation source (strict blinded judge, 5-year horizon).

Source	Addressed
Direct LLM	0.88
Author future-work	0.51
Evidence-tension	0.47
Weak/negative controls	0.32
Paper 1 evidence-graph ($n = 10$)	0.20

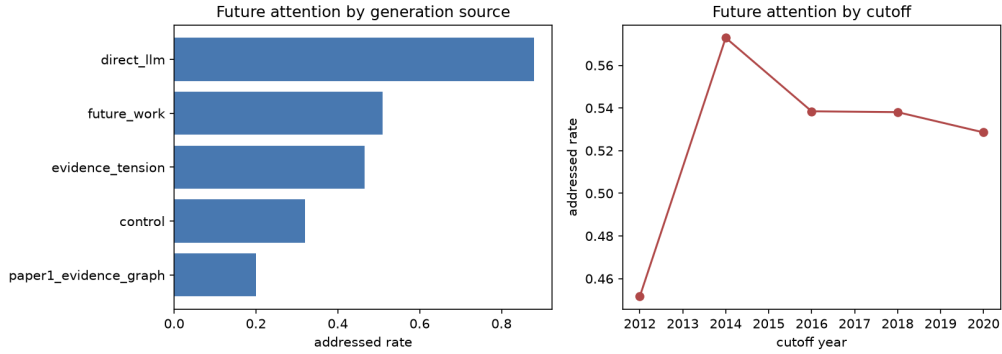


Figure 1: Addressed rates by generation source and cutoff.

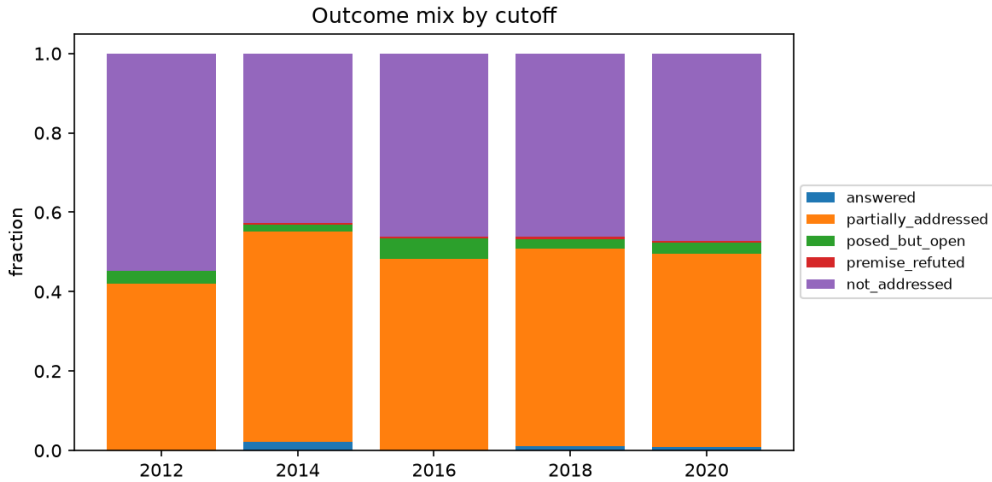


Figure 2: Outcome status mix across the 980 frozen questions.

Controls land lowest — the dataset’s sanity check passes. The direct-LLM rate of 0.88 should

be read with care: the generating model may know, parametrically, which 2016-era questions became hot topics, an irreducible leakage channel we document rather than claim to eliminate (Section 7). The mined sources (tension, future-work) are built from pre-cutoff sentences and cannot leak this way; that they land at ~ 0.5 against a strict judge is the honest base rate of real open problems. The ten Paper 1 questions — all engaged by future literature under Paper 2’s full-text, hand-curated evaluation — score only 2/10 here, a direct measurement of how much this paper’s abstract-only retrieval and strict direct-engagement bar *under-count* engagement for narrow, object-level questions (they average 3.4 entities per question vs. 1.5 elsewhere).

5.2 Predicting future attention (Task A)

Temporal split, test = cutoff 2020 ($n = 210$, base rate 0.53); results in Table 2.

Table 2: Task A (future attention) on the temporal test split.

Model	AUC	Notes
Majority class	0.500	
Field popularity only	0.519	popularity is <i>not</i> a question-level predictor
Question text only (BoW)	0.772	absorbs source style + topic hotness
Logistic (structured, L2)	0.713	interpretable
Logistic (L1)	0.707	
Random forest	0.718	
Gradient boosting	0.688	

Three observations. First, everything question-level beats field popularity by a wide margin: which questions succeed is not just which fields are hot. Second, raw question text is the strongest single signal — but it is also the least meaningful, because it can encode the generator’s stylistic fingerprint (direct-LLM questions are both stylistically distinctive and 88% addressed). Third, the structured features reach AUC 0.71 while being *auditable*: every unit of signal is a named, cutoff-time property.

Out-of-domain. Trained on four subfields and tested on four entirely unseen ones ($n = 470$), the structured logistic model holds AUC **0.710** (GBM 0.647). Leave-one-subfield-out AUCs span 0.62–0.78 with every subfield above chance (stellar activity 0.78, gravitational waves 0.75, cosmology 0.71, disks 0.70, exoplanet atmospheres 0.69, galaxy evolution 0.66, FRBs 0.66, compact objects 0.62). Question-level predictors are not a memorized property of any one community.

Ablations. (GBM, temporal test; Figure 3): evidence-context is the load-bearing group — dropping it costs the most ($0.688 \rightarrow 0.665$) and it alone nearly matches the full model (0.685). Text-only structure reaches 0.578, falsifiability 0.543, tractability 0.540, environment 0.550. Dropping the novelty group *helps* (0.708), foreshadowing the hypothesis results.

5.3 Which properties robustly predict attention

Controlled logistic effects (standardized, with subfield + cutoff + source + environment covariates), with sign-stability across the 5 cutoffs / 8 subfields / 4 sources, are given in Table 3 and Figure 4.

The picture that survives all four analyses:

1. **The community mostly answers questions it has already begun to recognize.**
Overlap with pre-cutoff reviews is the strongest, most stable predictor — future attention

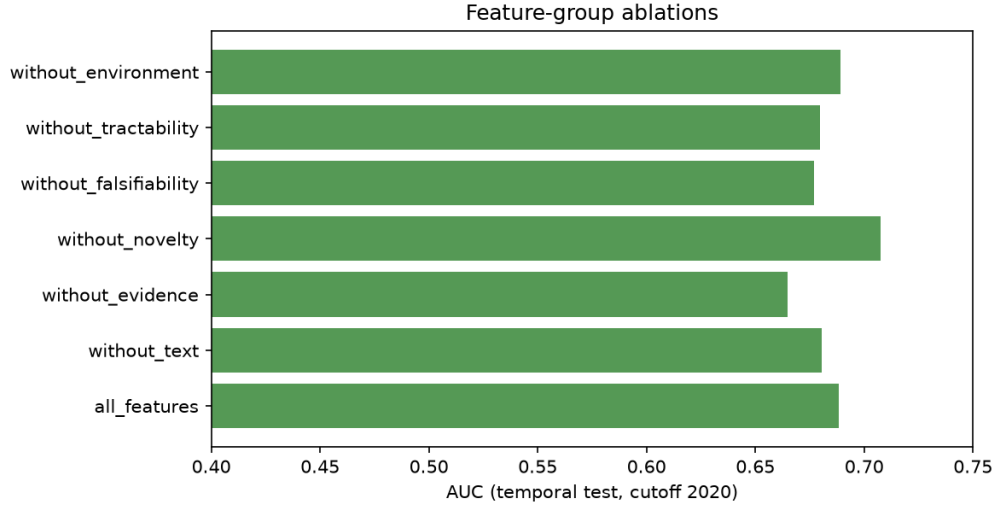


Figure 3: Feature-group ablations (GBM, temporal test split).

Table 3: Controlled effects on future attention with sign-stability fractions across cutoffs / subfields / sources.

Predictor	β	p	Sign-stable (cut./subf./src.)
Review overlap (already visible in pre-cutoff reviews)	+0.34	4×10^{-5}	1.00 / 0.88 / 1.00
N. directly similar prior papers	+0.26	0.005	0.80 / 1.00 / 0.75
Instrument diversity of prior evidence	-0.21	0.005	1.00 / 0.75 / 1.00
Explicit competing hypotheses (LLM-coded)	+0.17	0.021	0.80 / 0.88 / 0.50
Semantic novelty (distance to prior corpus)	+0.17	0.030	1.00 / 0.75 / 0.50
Entity count	-0.17	0.039	0.80 / 0.75 / 0.50
Specificity score	-0.16	0.041	0.80 / 1.00 / 0.50
Evidence tension density	+0.06	0.52	0.60 / 0.62 / 0.50

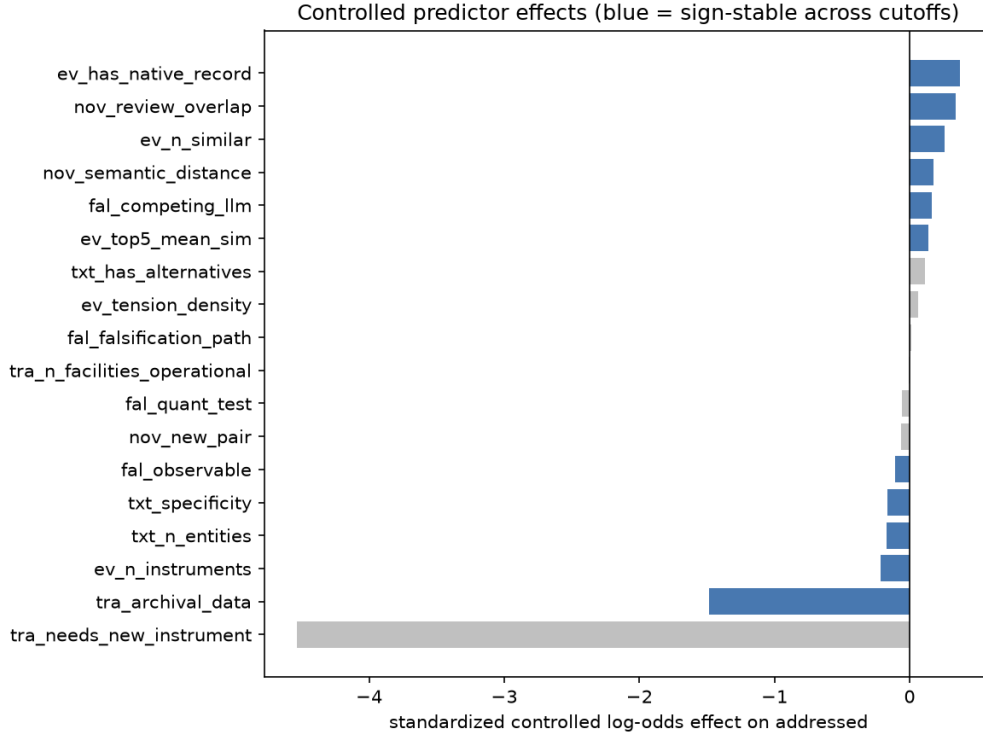


Figure 4: Controlled predictor effects on future attention.

is highly autocorrelated with present institutional attention, even after controlling for field size and growth.

2. **A dense base of directly related evidence predicts uptake; evidence *tension* per se does not.** What matters is that many papers already bear on the question, not that they disagree.
3. **Questions whose evidence spans many instruments are *less* likely to be engaged** — consistent with multi-instrument questions being synthesis-level problems that no single group owns.
4. **Explicitly articulated competing hypotheses help** — the one falsifiability property with predictive teeth.
5. **Specificity cuts against measured attention.** Highly entity-specific questions are engaged less often and *more slowly* (Cox HR 0.73 per SD, $p < 10^{-4}$) under literature-bounded labels — partly a real narrowness effect, partly the retrieval bound (Section 7).

Time-to-engagement (Task C). In the Cox model (516 events/980), retrieval-similarity mass dominates speed (HR 3.77 per SD), field growth accelerates engagement (HR 1.15), an explicit falsification path accelerates it (HR 1.15), and specificity (HR 0.73) and tension density (HR 0.88) slow it. Within-cell ranking of 2020 questions by model score correlates with realized future paper counts at mean Spearman $\rho = 0.29$ across the 8 test cells.

Outcome type (Task B). The five-way status classifier reaches macro-F1 0.23 (accuracy 0.56): it separates addressed/partial from not-addressed but cannot yet learn the rare classes (8 answered, 4 refuted in total) — exactly the sample-size regime the design anticipated, and the reason Task A is primary at $n \approx 1,000$.

5.4 Pre-stated hypotheses: mostly falsified

Table 4: Verdicts on the six pre-stated hypotheses.

Hypothesis	Verdict
H1 tension > novelty	No. Novelty carries a small positive controlled effect ($\beta = 0.17$, $p = 0.03$); tension density is null ($\beta = 0.06$, $p = 0.52$).
H2 observation availability $\rightarrow$ attention	No. Operational-facility count is null on attention ($p = 0.91$); refutation side untestable (4 cases).
H3 competing hypotheses $\rightarrow$ answered	Directionally yes, underpowered. On attention: $\beta = 0.17$, $p = 0.02$. On answered-given-addressed: $\beta = 0.56$, $p = 0.08$.
H4 novelty inverted-U	No. Quadratic term null ($p = 0.46$).
H5 popularity $\rightarrow$ counts, not answers	Untestable as stated. Count effect positive but n.s. ($p = 0.21$); answer side degenerate (8 answered).
H6 tension $\times$ tractability interaction	Marginal. $\beta = 0.16$, $p = 0.053$ — suggestive, unconfirmed.

We regard this table as a feature of the method, not a failure of the paper: five of six mechanism intuitions that sound compelling in a proposal do not survive contact with a thousand historical outcomes and proper controls.

5.5 Interpretable prediction cards

The released pipeline emits, for any frozen question, a card of the form:

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Question: How does the dark matter distribution in the innermost regions
of dwarf galaxies compare to the predictions of cold dark matter
simulations? [evidence_tension, cutoff 2020]
Predicted P(future attention): 0.50  $\rightarrow$  observed: addressed (partial)
Main positive factors: review overlap, directly similar prior papers,
explicit competing explanations
Main negative factors: high entity specificity, multi-instrument evidence
base

```

A cautionary card from the Paper 1 import: “*To what extent have actual JWST observations validated pre-launch predictions that aerosol-free TRAPPIST-1 CO₂ features should be detectable...*” (frozen at 2020-12-31, predicted 0.003, observed not addressed under our retrieval). The phrase “actual JWST observations” presumes data that did not exist at the cutoff — an instance of generation-time knowledge contaminating the *phrasing* of a nominally frozen question. Every LLM-mediated pipeline, including ours and Paper 1’s, needs phrasing audits of exactly this kind; the dataset makes such audits possible because provenance is stored verbatim.

6 Discussion

What the dataset says about question value. The strongest honest summary: five-year scientific attention is predictable to a useful degree (AUC ~ 0.7 out-of-time and out-of-domain) from cutoff-time properties, but the predictive properties are about the question’s *relationship to its community* — already-in-reviews, dense direct evidence, articulated alternatives — more than about the intrinsic virtues the literature on “good questions” celebrates (novelty, tension, specificity). Attention follows preparation. Whether that is how science *should* allocate attention is precisely the kind of question this dataset now lets others study: the rare premise-refuted cases, the slow-burning specific questions, and the unaddressed-but-well-formed tail are all released with full provenance.

Specificity and the measurement bound. The negative specificity effect is partly real (narrow questions have small addressable communities) and partly instrumental: abstract-level TF-IDF retrieval misses engagement with object-level questions, as the Paper 1 subset demonstrates (2/10 here vs. 10/10 under Paper 2’s full-text evaluation). We report it as “less *measured* attention”, never “worse questions” — and this is the main reason Task A labels should be read as lower bounds.

The direct-LLM anomaly is a warning, not a victory. The 88% addressed rate for direct-LLM questions is exactly what parametric leakage would produce: a model asked in hindsight for “the most important open questions of 2016” can simply recall what 2017–2021 worked on. Because source is a covariate in every regression, the predictor analysis is shielded from this; but any future benchmark that scores generators by addressed-rate alone will be gamed by hindsight knowledge. Mined sources with verbatim pre-cutoff provenance are the defensible foundation.

7 Limitations

- **Abstract-level corpus.** Full texts are not used; tension and future-work mining see only abstracts, and citation-weighted attention is unavailable without a second data provider. Paper counts, lead times, and review recognition stand in for citation impact.
- **Retrieval-bounded labels.** A question can be engaged by literature our TF-IDF retrieval misses; not_addressed is a lower bound on engagement, uniformly across sources but not uniformly across specificity (see Section 6).
- **LLM judges, not humans.** Label reliability is estimated with an independent second model rather than human adjudication (63.5% raw agreement, $\kappa = 0.21$, disagreement one-directionally lenient). The annotation protocol, rationales, and supporting ids are released so human passes can replace or audit the judges; per-class precision tables are in the released agreement report.
- **Parametric-knowledge leakage.** The direct-LLM source (and, weakly, LLM phrasing of mined material) can import post-cutoff knowledge despite freeze instructions — including into question phrasing, as the JWST card shows. Source is controlled for in all analyses; the direct-LLM addressed rate should not be read as generator quality.
- **Rare outcomes remain rare.** 8 answered and 4 premise-refuted questions cannot support the Task B taxonomy or H2/H5 as stated; the design document’s projection that these classes need 2,000–5,000 questions is confirmed empirically.
- **One domain.** Astronomy only; the cross-subfield transfers here are necessary but not sufficient evidence of cross-science generality.

8 Conclusion

We built the first temporally grounded, multi-cutoff, multi-source dataset of scientific questions with future-outcome labels, and used it to replace two common practices: scoring questions by how they sound, and validating question generators on single-digit case studies. At $n = 980$ with strict blinded labeling, future scientific attention is predictable ($AUC \approx 0.71$ out-of-time and out-of-domain) — and the robust predictors are community-relational, not rhetorical. Most pre-registered mechanism hypotheses failed, which is the point: a dataset an order of magnitude larger than its predecessor makes intuitions testable, and most did not survive. The dataset,

every prompt, every judge rationale, and the full pipeline are released for the community to reuse, audit, and extend to other sciences.

Reproducibility

All code, the frozen dataset, labels, and results are released; every pipeline stage is a single script, all LLM calls are cached and pinned to temperature 0, and the offline test suite plus a temporal-isolation validator run in CI. Rebuilding the corpus from the public arXiv API and regenerating the dataset reproduced the frozen questions byte-for-byte.